

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2019



CE 2107
(Construction Technology and Materials)

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is sand? Write down the classification of sand accordingly to source. In which sand do you recommend for making cement concrete? (10)
(b) What is "Fineness modulus"? Explain its significance in sand grading. (05)
(c) What is bulking of sand? Draw the qualitative gradation curve for well graded, gap graded and uniformly graded sample. (10)
(d) A mixed sample of 1200gm of two different variety of sands, one from Savar having Fineness modulus = 2.20 and the other from Sylhet having Fineness modulus = 2.80. If the combined fineness modulus was to be 2.55; calculate the amount of each variety of sand in the mix. (10)
2. (a) What is brick? Give the constituents of good brick clay and state their functions. (10)
(b) What is frog mark? What are the tests that are carried out in the field to determine the quality of good bricks? (08)
(c) Draw the plan view of Hoffman's kiln. Explain also its working principle. (12)
(d) What is FRP? Write down the uses of FRP. (05)
3. (a) What is cement? Discuss the importance of cement in engineering constructions. (08)
(b) Write down the differences between false and flash setting. (06)
(c) What are the different methods of manufacturing ordinary Portland cement? Explain the different steps in the dry process of manufacturing Portland cement with a neat flow diagram. (15)
(d) Write down the differences between bleeding and laitance. (06)
4. (a) What is concrete? Discuss the factors controlling the quality of a concrete. (10)
(b) What is segregation of concrete? Write down the causes and effects of segregation. How to remedy the segregation of concrete? (13)
(c) Write short notes on: (12)
 - i. Workability of concrete;
 - ii. Creep of concrete;
 - iii. Slump test
 - iv. Curing of concrete.

Section – B

5. (a) What is foundation? Write down the purposes of foundation. (07)
- (b) When deep foundation is provided? Write the classification of pile foundation according to their load transmission and functional behavior. (08)
- (c) Write short notes on: (i) Pre-cast; and (ii) Cast in-situ concrete pile. (08)
- (d) Draw the neat sketch of the following foundations. Also state why and when they are best they are best suited for – (12)
- (i) Combined footing; (ii) Steeped foundation; (iii) Pile foundation
6. (a) What is stair? Discuss briefly the requirements of a good stair. (10)
- (b) Define the following terms: (06)
- (i) Winders; (ii) Steps; (iii) Newel post
- (c) "Reinforced concrete stairs predominate the stairs made from other material". Justify the statement. (08)
- (d) Write down the name of different types of stairs with their neat sketch. (11)
7. (a) What are the components of flooring? What are the functional requirements of floor? (15)
- (b) What is roof? Briefly describe the types of pitched roof. (10)
- (c) Briefly describe the following terms: (10)
- (i) Framed and paneled door; (ii) Louvered door; (iii) Double-hung windows; (iv) Slider; (v) Skylight
8. (a) What are objects of plastering? Describe the defects found in plaster. What is Brick nogging partitions? (15)
- (b) What are the components of plumbing? Draw the schematic diagram of Domestic incoming Electrical service. (10)
- (c) What are the facilities of segmental construction technique in Bridge construction? (05)
- (d) Shortly describe the construction procedure of concrete pavement. (05)
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Khulna University of Engineering & Technology
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BURP 2nd Year 1st Term Regular Examination, 2019

Hum 2117
(Sociology for Planners)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define Sociology. Discuss the origin & development of sociology. (10)
(b) Explain the importance of sociology for the students of URP. (15)
(c) Describe the elements of social structure. (10)
2. (a) Describe different types of social structure in different society. (15)
(b) Describe the Marx's concept of social structure with example. (10)
(c) What is called society? Mention the major characteristics of society. (10)
3. (a) What do you understand by community? (05)
(b) Distinguish between society and community with example. (15)
(c) What is family? Describe the impacts of globalization in changing role of family in Bangladesh. (15)
4. (a) What do you mean by culture? (05)
(b) What are the elements of culture. (10)
(c) Distinguish between culture and civilization with example. (10)
(d) What is socialization? Discuss the role of family for socialization. (10)

Section – B

5. (a) Define family. Discuss the major functions of family. (10)
(b) What is juvenile delinquency? Describe the causes, effects and remedies of juvenile delinquency. (15)
(c) Find out the changing dynamics of urbanization in developing countries. (10)
6. (a) What is urban ecology? (05)
(b) What are the effects of ecological elements on social life? (15)
(c) Define crime. Find out the major causes of crime in urban area. (15)
7. (a) Define power. (05)
(b) Write down the leadership pattern in our country. (15)
(c) Show the influence of globalization in the development of our country. (15)
8. (a) Define poverty. Explain the major causes of poverty in Bangladesh. (10)
(b) Define social stratification. Why human society is stratified? (15)
(c) Briefly discuss the culture of Bangladesh. (10)

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URP 2113
(Site and Area Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

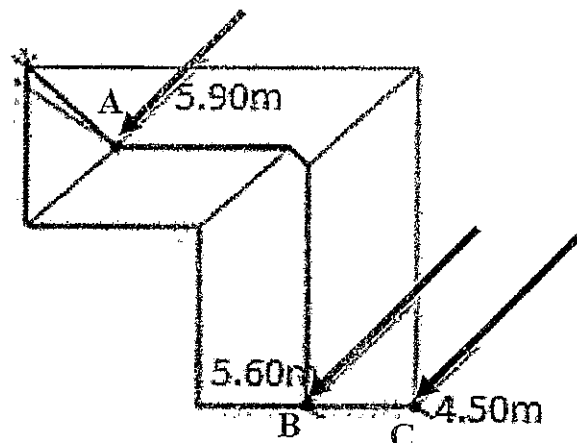
1. (a) What is meant by Site Planning? Write down the scope of site planning. (07)
(b) Provide appropriate diagrams to portray Kevin Lynch's site planning process. (10)
(c) "Site planning is related to spirit of the place, character of the place, nature of the project and behavioral studies" – Justify the statement with your rationalities by providing pertinent examples, if necessary. (10)
(d) "Site planning should be done in connection with the nature" – Narrate the statement. (08)
2. (a) "Preparation of site inventory helps to do site analysis that includes various elements of a site". Give a short overview of the various elements of a site inventory checklist. (15)
(b) What is slope? What does the 7% mean? (05)
(c) Write down the meteorological, physical and human factors affecting surface runoff. (15)
3. (a) Write down the importance of parks and open spaces policies in Bangladesh. (10)
(b) How can land use controls achieve ecological sustainability? (15)
(c) Explain the concept of sun path diagram with necessary sketches. Why is it important to know the sun path diagram in site planning? (10)
4. Write short notes on the following issues (any five) (35)
[Note: Provide appropriate diagrams wherever necessary]
(a) Classifications of parks and open space with examples
(b) Textural triangle
(c) S.M.A.R.T strategy
(d) Burgess zone model
(e) Climate elements
(f) Zoning

Section – B

5. (a) "Site planning is involved with several professions viz. Urban Planning, Architecture, Landscape Architecture and Civil Engineering"- Briefly explain. (06)
(b) Access points to public streets should be limited to minimize interference. Why do you think so? (07)
(c) Recall some practical example of bad design in planning concept. Critically discuss with the solution (Hint: Figures are mandatory) (12)

- (d) In site planning, road network are classified into four categories (major thoroughfare, major distributor, minor distributor and access roads) based on the functions. Clarify the statement. (10)
6. (a) Prepare a shadow diagram at 9.00 am for point A, B and C of the plan from the given information: (20)

Chart 1		Chart 2	
Time	Angle (Azimuth)	Time	Length of Shadow Cast by 1m Pole on a Flat Land
9.00 am	46°25'	9.00 am	3.56 m
12.00 pm	5°52'	12.00 pm	2.25 m
3.00 pm	325°25'	3.00 pm	2.15 m



- (b) Suppose you are assigned to make an inventory checklist for analyzing a specific site. What will be your considerations? (05)
- (c) Mention some factors for the design of the street system with proper description. (10)
7. (a) Why lot design is a critical component? Write about the benefits of using technologies for site and area planning in the modern era with its limitations. (15)
- (b) Write down the general conditions for the approval of a private residential housing project under the Private Residential Housing Project Land Development Rules, 2012. (10)
- (c) Draw a differentiating table between Grid Iron and Modified Grid Iron pattern. (10)
8. (a) Write down the objectives, performance criteria and prescriptive measures with proper figure and explanation of the following elements of residential subdivision planning. (21)
- (i) Density (ii) Allotment Design (iii) Landscaping
- (b) Annotate the following pattern of subdivision. (10)
- (i) Combination pattern (ii) Loop Street Pattern.
- (c) Site plans locate objects and activities in SPACE and TIME. Why? (04)

Khulna University of Engineering & Technology
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BURP 2nd Year 1st Term Regular Examination, 2019.
URP 2115
(Urban Planning Techniques)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Write about the concept of "Urban Development Plan" in controlling unplanned urbanization of Bangladesh. (10)
(b) Clarify the following components of urban development plan with proper example. (10)
(i) Equity in development
(ii) Formulation of policy and regulation for development control
(c) Mention some common characteristics of Urban Development Plan for ensuring sustainable development of a city. (10)
(d) Illustrate the forms of urban development plan and mention which one is not currently practiced in Bangladesh? (05)
2. (a) Why master plan is called a comprehensive plan? Master Plan is not practiced at present in our country because of some difficulties to apply – specify those difficulties with example. (10)
(b) Make a comparative analysis between Strategic Plan and Structure Plan with proper example. (15)
(c) "Detailed Area Plan is currently practiced to overcome some problems of Master Plan". Make a summary on those features of Detailed Area Plan. (10)
3. (a) Suppose, you are working as a town planner in Khulna Development Authority and going to develop the fringe area (residential) of the city. Which technique will you choose between Guided Land Development and Land Readjustment Plan? Why? (15)
(b) Critically analyze Land Banking technique of participatory area planning. (10)
(c) "Area Improvement Plan is feasible in already developed areas of a town" – do you agree? Discuss the methods of development used in this technique. (10)
4. (a) Write short notes as the following directions of Dhaka Metropolitan Building Construction Rule '2007. (20)
(i) Height of building (ii) Parking Provision (iii) Necessary arrangement of roads (iv) Water supply, sewerage and drainage
(b) "Violation of some standards regarding fire safety in building are responsible for devastating event of disruption and casualties" – discuss in light of recent two incident of fire hazards in Dhaka (February). (15)

Section – B

5. (a) Define population "Estimation". Differentiate between "Projection" and "Forecast". How population projections are often mislabelled as forecast? (10)
(b) Differentiate between Gompertz curve and Logistic curve in some points. (10)
(c) From the following data which curves give the best possible fit for region A and region B? – explain the reason. (10)

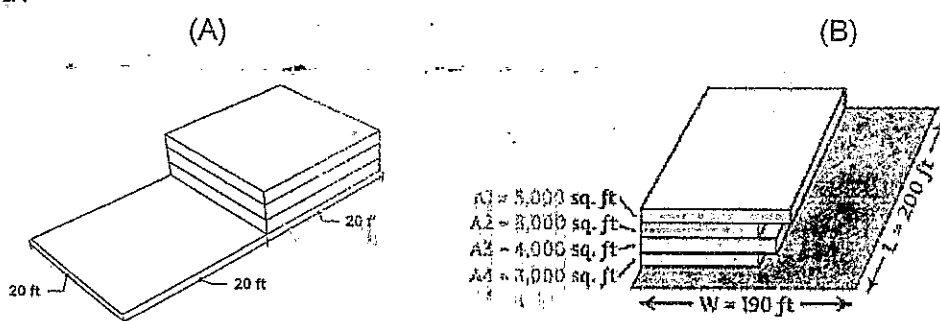
Observation Period	Population (A)	Population (B)
0	20,000	20,000
1	21,500	30,000
2	23,000	45,000
3	24,000	67,500
4	26,000	101,250
5	27,500	151,875

(d) Discuss the assumptions of Location Quotient Approach. (05)

6. (a) Define basic sector and non-basic sector with examples. (05)
 (b) Define Base Multiplier and Location Quotient Approach, with necessary equations and meaning of symbols. Mention the limitations of Location Quotient Approach. (10)
 (c) Show the Conceptual Framework for housing choice. (08)
 (d) If the reference region growth rate is constant, what will be the employment for "C" in 2000 with the reference year of 1985? (12)

Industry	Employment		
	1983	1985	2000
A	133.4	5240	125.4
B	19.7	120	30.9
C	27.5	913	37.2

7. (a) What do you mean by "Consistency" and "Change" in land use map? Discuss about the categories of growth areas. (10)
 (b) Describe the steps of suitability Analysis taking consideration that is for agricultural zone. (15)
 (c) Calculate the FAR for both A and B. Which one is suitable for high density area? (10)



8. (a) What do you mean by Recreation centres, Parks and Open spaces (RPOS)? Mention the points to be considered for designing RPOS. (10)
 (b) Differentiate between renewal and demolitions in urban area. Describe the dimensions of urban renewal. (15)
 (c) Suppose the new population of a city is 39,146 where proportion of persons living in group homes is 3.11% and weighted average of household size is 2.5. Then what will be the number of total new housing units in that city? Consider the following data: (10)

Assumptions and values	Single family	Multiple family
Housing mix	75.36%	24.64%
Household size	2.7	1.9
Vacancy rate	4.52%	12.75%

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2019
URP 2191
(Statistics for Planners-I)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.
iii) Necessary graphs and tables will be supplied.

Section – A

1. (a) Define Statistics and Statistical Methods. Mention the functions of statistics in short. (10)
- (b) Explain different levels of measurement of statistical data with appropriate examples. (10)
- (c) Draw Histogram; Frequency polygon and Ogive from the following data: (15)

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	5	10	15	8	17	9	3

2. (a) Write down the key criteria for determining number of classes, class intervals and class limits. When should we use classes with undefined limits and class intervals? (10)
- (b) What are the key components of a good table? The individual output of 60 female workers of an industrial firm in one week is given below: (15)

123	78	137	111	156	198	102	186	234
145	96	149	126	178	193	107	198	246
156	67	155	142	187	180	119	195	228
167	87	167	154	173	165	127	192	234

Present the above data in the form of a frequency table using 12 class intervals.

- (c) Below data show the variation of a share price (highest and lowest) during the last twelve months. Draw appropriate curve showing the variation of the share price during each month. (10)

Month	Highest Price Per Share (Tk.)	Lowest Price Per Share (Tk.)
Jan	135	120
Feb	140	127
Mar	140	135
April	145	137
May	150	140
June	150	142
July	158	145
Aug	165	150
Sep	168	154
Oct	170	155
Nov	160	150
Dec	156	140

3. (a) What are the differences between Arithmetic Mean and Geometric Mean? What are the applications of Geometric Mean? (07)
- (b) Mean, median, mode and standard deviation- all of these are necessary to have a good idea about any dataset. Explain with examples. (09)
- (c) Locate median through Ogive Curve from the following table showing the percentage distribution of households in a rural area of Bangladesh according to (09)

income.

Income Group	Percentage of HH
Upto 499	3.0
500-999	24.2
1000-1499	30.2
1500-1999	17.0
2000-2499	9.4
2500-2999	6.0
3000-3499	4.9
3500-3999	2.2
4000-4499	3.1

- (d) If $\bar{x}$ and σ denote respectively mean and standard deviation of a set of n non-negative observations, then prove that $\bar{x}\sqrt{1-n} \geq \sigma$ (10)

4. (a) A sample of daily production was organized into the following distribution: (15)

Daily Production	Frequency
80-90	11
90-100	09
100-110	15
110-120	20
120-130	08

Calculate: (i) Co-efficient of Variation, (ii) Mean Deviation about Median

- (b) What are the differences between Skewed Curve and Normal Curve? Below table shows income distribution in percentage of Norway and Bangladesh. (10)

Norway		Bangladesh	
Percentage of Population	Percentage of Income	Percentage of Population	Percentage of Income
10	10	10	5
15	12	15	10
20	15	20	10
20	15	20	15
20	15	20	15
10	18	10	15
5	15	5	30

Which Country's income distribution is the most Skewed? How will you interpret such distribution?

- (c) Compute 6th Decile and 78th Percentile from the following dataset. (10)

Daily Wage in Tk.	No. of Workers
12.5-17.5	32
17.5-22.5	52
22.5-27.5	79
27.5-32.5	114
32.5-37.5	93
37.5-42.5	64
42.5-47.5	26
47.5-52.5	11
52.5-57.5	07

Section – B

5. (a) Define with example (any four) (16)
- (i) Sample space; (ii) Event; (iii) Dependent and independent event; (iv) Mutually exclusive event; (v) Conditional probability.
- (b) A firm conducted a survey to study whether changes in office design increase productivity. 70% workers said that reducing noise would increase productivity. 67% workers said that more storage space would increase productivity. 56% workers said that both noise reduction and increased storage space would improve productivity. If a worker is randomly selected (09)

and asked what changes in office design would increase worker productivity, what is the probability that this person would select reducing noise or more storage space?

- (c) According to a survey, the probability that a family owns a car if its monthly household income is greater than BDT 100000 is 0.75. Of the households surveyed, 60% had incomes over BDT 100000 and 52% had a car. A family is randomly selected; Find out the probability that the family has a car and an income over BDT 100000 monthly. (10)

6. (a) A candidate is selected for interview of management trainees for 3 companies. For the first company there are 12 candidates, for the second there are 15 candidates and for the third there are 10 candidates. What are the chances of his getting job at least at one of the companies? (05)

- (b) The Human Resource Department of a company has records which show the following analysis of its 200 planners. (08)

Age	Bachelor's Degree	Master's Degree	Total
Under 30	90	10	100
30-40	20	30	50
Over 40	40	10	50
Total	150	50	200

If one planner is selected at random from the company, find:

- (i) the probability that the planner has only a bachelor's degree.
(ii) the probability that the planner has a master's degree, given that the planner is over 40.
(iii) the probability that the planner is under 30, given that the planner has only a bachelor's degree.

- (c) In a town, it's rainy one third of the days. Given that it is rainy, there will be heavy traffic with probability $1/2$, and given that it is not rainy, there will be heavy traffic with probability $1/4$. If it's rainy and there is heavy traffic, Mr. X arrives late for work with probability $1/2$. On the other hand, the probability of being late is reduced to $1/8$ if it is not rainy and there is no heavy traffic. In other situations (rainy and no traffic, not rainy and traffic) the probability of being late is 0.25. Now, draw a tree diagram from the above data and pick a random day. (12)

- What is the probability that it's not raining and there is heavy traffic and Mr. X is not late?
- What is the probability that Mr. X is late?
- Given that Mr. X arrived late at work, what is the probability that it rained that day?

- (d) A committee of 5 persons is to be formed from 6 men and 4 women. In how many ways this can be done when (10)
- at least 2 women are included?
 - at most 2 women are included?

7. (a) Define Bernoulli Population and Bernoulli trial? Mention the characteristics of a Binomial Probability Distribution. (12)

- (b) Assume that on an average one telephone number out of fifteen is busy. What is the probability that if six randomly selected telephone numbers are called (12)
- not more than three will be busy?
 - at least three of them will be busy?

- (c) The number of flaws in a fibre optic cable follows a Poisson Distribution. The average number of flaws in 50m of cable is 1.5 (11)
- What is the probability of exactly three flaws in 200m of cable?

- ii. What is the probability of at least two flaws in 100m of cable?
 - iii. What is the probability of exactly one flaw in the first 50m of cable and exactly one flaw in the second 50m of cable?
8. (a) What is meant by Standard Normal Distribution? Mention the properties of a Normal Probability Distribution. (12)
- (b) As a result of tests on 20,000 electric fans manufactures by a company, it was found that life time of the fans was normally distributed with an average of 2,040 hours and standard deviation of 60 hours. On the basis of the information, estimate the number of fans that is expected to run for (i) more than 2,150 hours and (ii) less than 1,960 hours. (11)

[Note: Probabilities should be shown graphically]

- (c) The income of a group of 10,000 persons was found to be normally distributed with mean tk. 1750 per month and standard deviation of tk. 50. Show that of this group 95% had income exceeding tk. 1668 and only 5% had income exceeding tk. 1832. What was the lowest income among the richest 100? (12)

[Note: Probabilities should be shown graphically]